Climate Change Analysis

Name: Jency Francis

Student ID: 23082639

1.Introduction

Climate change is a pressing concern defined by shifts in weather patterns and the alarming rise in global temperatures, primarily driven by human activities such as deforestation and the combustions of fossil fuels.To gain a deeper insight into this critical topic, a dataset on climate change has been selected.

2.Important Statistics

The table summarizes the key statistics obtained from the dataset:

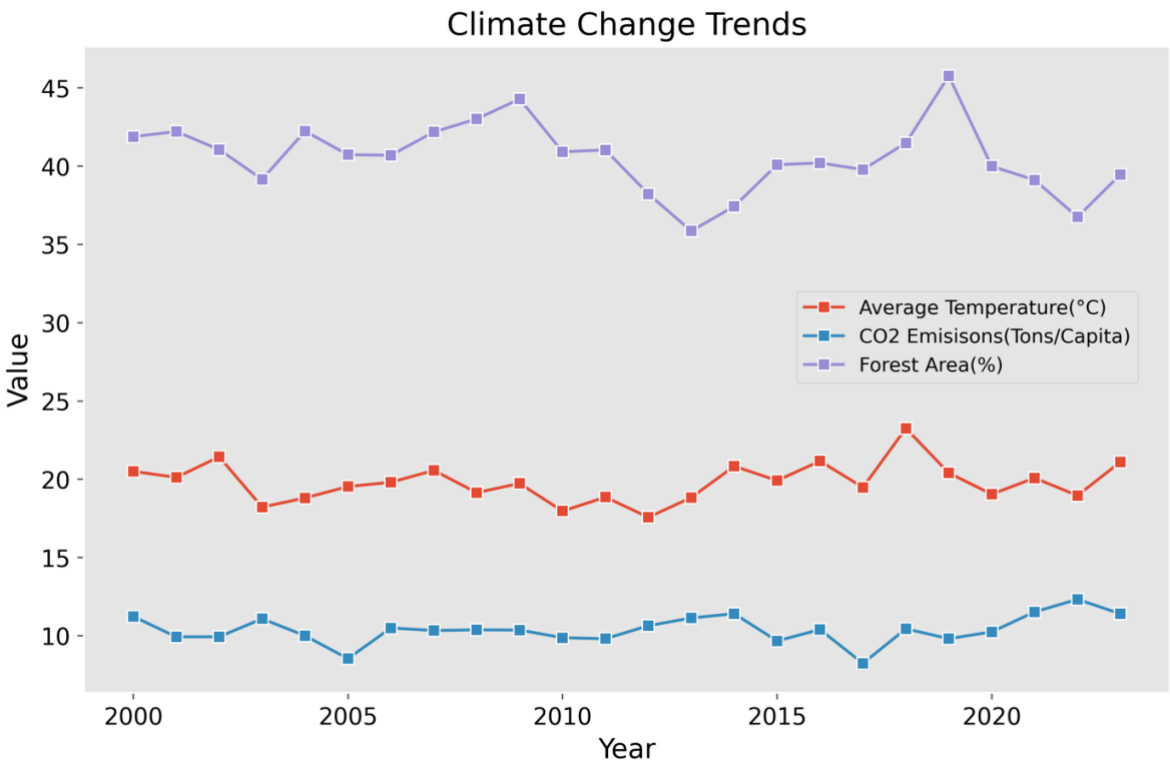
Basic Statistics

	Year	Avg Temperature (°C)	CO2 Emissions (Tons/Capita)	Sea Level Rise (mm)	Rainfall (mm)	Population	Renewable Energy (%)	Extreme Weather Events	Forest Area (%)
count	1000.00	1000.00	1000.00	1000.00	1000.00	1000.00	1000.00	1000.00	1000.00
mean	2011.43	19.88	10.43	3.01	1738.76	705383000.00	27.30	7.29	40.57
std	7.15	8.54	5.61	1.15	708.98	409391000.00	12.97	4.42	17.40
min	2000.00	5.00	0.50	1.00	501.00	3660891.00	5.10	0.00	10.10
25%	2005.00	12.18	5.58	2.00	1098.75	343624200.00	16.10	3.00	25.60
50%	2012.00	20.10	10.70	3.00	1726.00	713116600.00	27.15	8.00	41.15
75%	2018.00	27.23	15.40	4.00	2362.50	1073868000.00	38.93	11.00	55.80
max	2023.00	34.90	20.00	5.00	2999.00	1397016000.00	50.00	14.00	70.00

3.Analysis and Findings

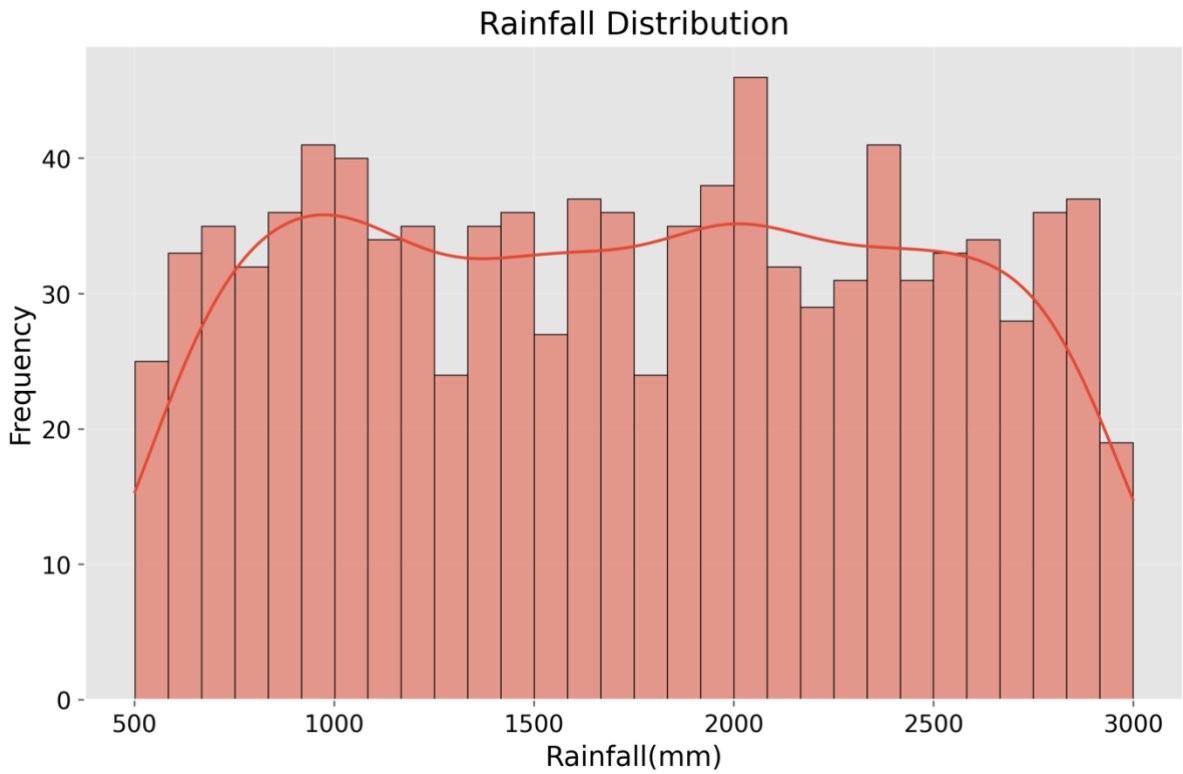
3.1.Climate Change Trends

A line plot has been created featuring three key factors such as average temperature,CO2 emissions, and forest area. As shown in the table, the **mean value** of average temperature, CO2 emissions and forest area are **19.88,10.43** and **40.57** respectively. There are fluctuations in the average temperature with a sudden drop in **2003** and an increase in **2018**.The line depicts an **increasing trend** over the last few decades.This rise in average temperature can be correlated with increasing CO2 emissions.It can be said that CO2 emissions are in control since there are no sharp peaks, while there are drops in **2005** and **2017** and shows a **steady trend** when compared to the average temperature. For forest area, there are evident fluctuations with equal amounts of ups and downs, where a major drop is seen in **2013** and the highest peak in **2019** and records a **stable trend**.



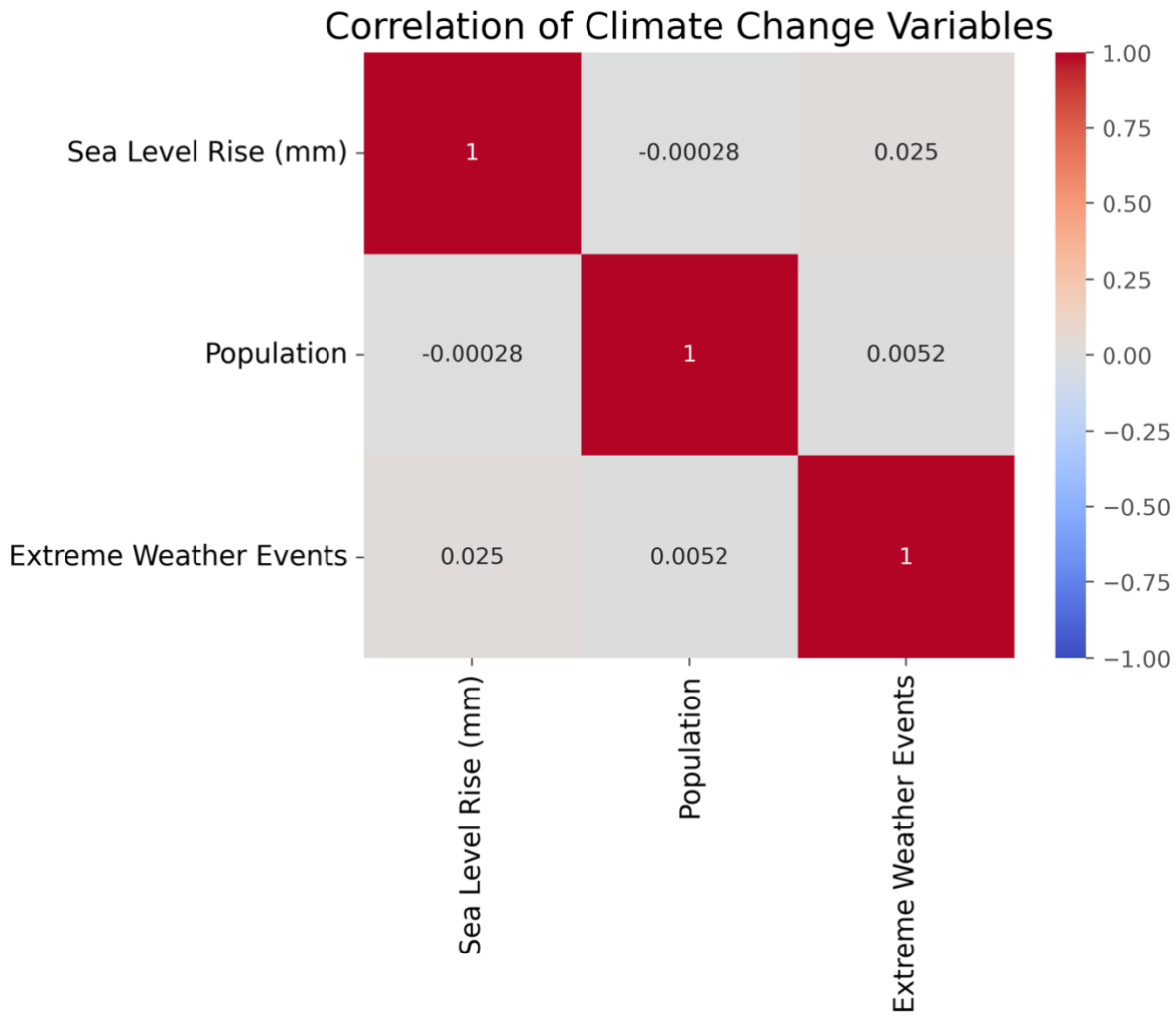
3.2.Rainfall Distribution

The histogram created on rainfall renders a **uniform distribution** with a positive **skewness** of **0.017** and is **platykurtic** with a value of **-1.21**.The **KDE line** shows that the data is wide spread with almost **same density** and **standard deviation** of **708.98** as seen in the table with **2000mm** rainfall in most of the years and **3000mm** being the lowest across the years.



3.3.Correlation of Climate Change Variables

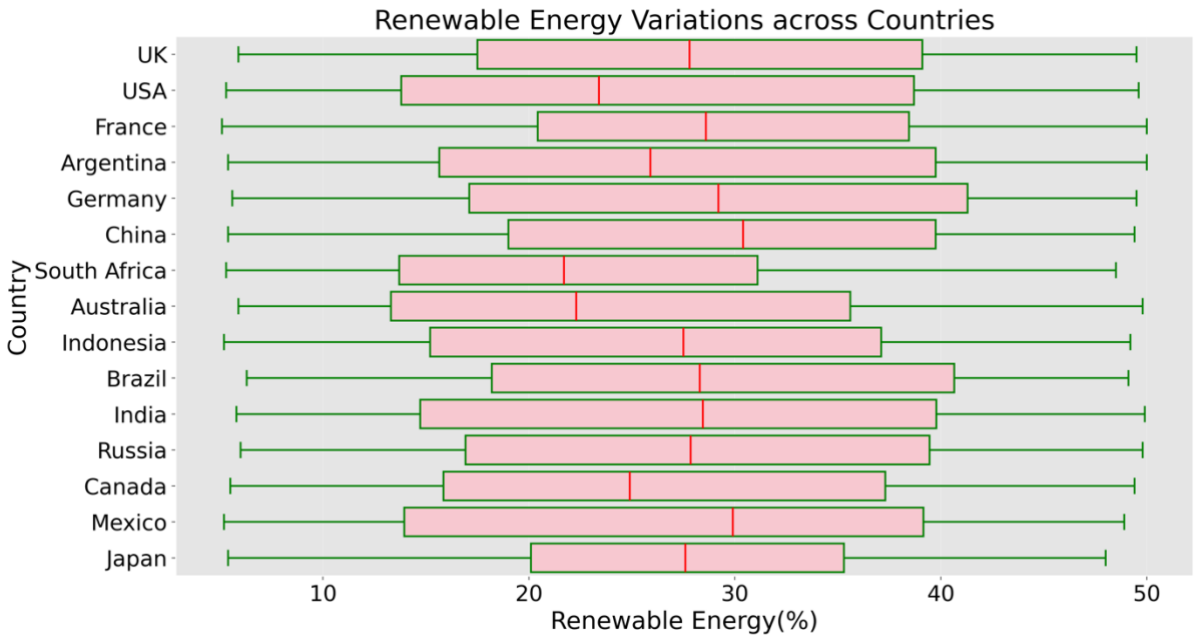
It can be implied from the heatmap that there is **no intense relationship** between the sea level rise, population and extreme weather events since the correlation coefficient values amongst these variables are very minute.The Sea level rise has a **positive correlation** of **0.025** with extreme weather events and a negligible **negative relationship** with population.Similarly, extreme weather events holds a **positive correlation** with population which is yet very weak and can be ignored.



3.4.Renewable Energy Usage across Countries

It can be stated from the box plot on renewable energy that there are no outliers. The **mean** and **standard deviation** are **27.3** and **12.97** respectively as shown in the table and **median** and **variance** are **27.15** and **168.24** respectively.Nonetheless there are some remarkable difference in renewable energy usage amongst the countries.Both China and Mexico have the highest median of slightly higher than 30% and 30% respectively, which implies that these countries are leading in the usage of renewable energy.Though, the widest interquartile range is seen for Mexico, the smallest belongs to South Africa.It also holds the lowest median which is slightly greater than 20%, which suggests that renewable energy usage is the lowest in South Africa amongst these countries.A narrower interquartile range has been observed for Japan recommending the consistency in renewable energy

usage.Moreover, a crucial variation can be noted for countries like Germany and France where significantly wide interquartile range is exhibited.



4.Conclusion

The analysis on climate change clearly outlines several key factors, including CO2 emissions,average temperature,rainfall distribution,sea level rise,and forest area.While there has been a consistent trend in reducing CO2 emissions and preserving forest area, many countries still fall short in their use of renewable energy.Additionally, addressing the challenges in controlling average temperature and enhancing renewable energy usage remain significant tasks to alter the adverse impacts of climate change.